# MANIK GOYAL

*Full Stack Software Engineer*

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## PROFESSIONAL SUMMARY

Full Stack Software Engineer with 2+ years building high-performance distributed systems in **C#/.NET 8, Angular and TypeScript** for a video analytics platform running on **400+ cameras and 200+ servers**; scaled sustained event throughput **16x**, cut report runtime **92%**, and shipped cross-platform **Flutter** apps and **LLM-powered engineering automation** alongside the core platform.

## TECHNICAL SKILLS

**Languages:** C# (C Sharp), Java, TypeScript, JavaScript, SQL, Dart, C++17, HTML5, CSS3

**Backend & Frameworks:** .NET 8/7, ASP.NET Core, Entity Framework Core, LINQ, SignalR, Spring Boot, REST APIs, Microservices, Event-Driven Architecture, Distributed Systems

**Frontend & Mobile:** Angular 18, React, Angular Material, RxJS, NPM package authoring, Component Libraries, Flutter (iOS & Android)

**Messaging & Streaming:** RabbitMQ, Apache Kafka, MQTT, Pub/Sub, Dead-Letter Queues, SignalR Hubs, WebRTC, FFmpeg

**Databases & Caching:** PostgreSQL, TimescaleDB, Redis, Milvus (vector database), SQL Query Optimization, Indexing, Partitioning, Compression

**Cloud & DevOps:** AWS (Amazon Web Services), Microsoft Azure, Docker, Kubernetes, GitHub Actions, CI/CD, Linux, Windows, Grafana, Prometheus, Observability

**AI & Automation:** LLM Workflow Automation, Agentic AI, Vector Embeddings, Similarity Search, Prompt Engineering

**Practices:** System Design, Clean Architecture, SOLID, Design Patterns, API Design, Performance Tuning, Code Review, Agile/Scrum, Unit Testing

**Spoken:** English (Professional), Hindi (Native), German (Basic)

## PROFESSIONAL EXPERIENCE

**Software Engineer, I2V Systems Pvt. Ltd.** Oct 2024 - Present

*Gurugram, India – .NET, C#, Angular, Flutter, RabbitMQ, PostgreSQL, Redis, Milvus, Docker*

- Led development of the Analytics Manager service in **.NET** for a **400+ camera** video analytics platform, coordinating ANPR, face recognition and intrusion events across **8+** backend microservices.
- Diagnosed message-processing bottlenecks and redesigned **RabbitMQ** consumption around event-specific queues with dead-letter handling, lifting sustained throughput from **30 to 500+ events/sec (16x)** while isolating failed messages for replay.
- Architected reusable messaging and storage abstraction libraries spanning **RabbitMQ, MQTT, local disk, NAS and cloud object storage**, adopted by **4+ services** and cutting new-service integration from days to hours.
- Optimized PostgreSQL and TimescaleDB ingestion with hypertables, native compression, retention policies and composite indexes, raising camera-frame processing from **70 to 300+ frames/sec**; tuned analytics indexes to cut query latency by **35%**.
- Implemented **Milvus** vector-embedding similarity search for face enrollment, watchlists and deduplication, reducing vector-index memory consumption by **75%** while preserving match accuracy.
- Reengineered analytics exports over **100K+ records** with streaming reads and incremental batch processing, cutting execution time from **12 minutes to under 1 minute (92% faster)** and eliminating out-of-memory failures.
- Built bulk enrollment workflows for **5,000+ records** with server-side validation and live **SignalR** progress streaming; integrated government e-challan REST APIs into the ANPR violation pipeline.
- Published **10+ reusable Angular/NPM component libraries** consumed by web apps across a **200+ server** deployment serving **60+ concurrent users**, standardizing UI behavior across products.
- Shipped cross-platform **Flutter (iOS & Android)** features with real-time alert delivery and background services for the surveillance product suite.
- Built **LLM-powered engineering automation**, including agentic PR review workflows, and automated Docker release pipelines with GitHub Actions, cutting release turnaround by **40%**; instrumented services with Grafana and Prometheus dashboards and alerts.
- Mentored **2** junior engineers on Clean Architecture and asynchronous C# patterns, and led structured code reviews that reduced QA-reported regressions.

## PROJECTS

**ParkNest, Peer-to-Peer Parking Marketplace (Live Beta)** 2025 - Present

*.NET 8, ASP.NET Core, PostgreSQL/PostGIS, Angular 18, Flutter, Docker – parknest.onrender.com · github.com/Manik400/ParkNest*

- Engineered and launched a two-sided marketplace on **ASP.NET Core**, now in **live beta in Gurugram**, supporting host listings, time-window renter bookings and prepaid credit settlement over a normalized PostgreSQL schema.
- Designed optimistic-concurrency booking safeguards and an idempotent, double-entry credit ledger (hold, release, overstay billing, settlement, payout) with row-locked wallets, preventing double-booking and duplicate charges under overlapping and retried requests.
- Built radius-based parking search on **PostgreSQL/PostGIS** with GiST-indexed geography queries and a pluggable memory/Redis cache, surfaced through an **Angular 18** admin and host console and a **Flutter** renter app with map search, QR-code check-in and live **SignalR** session updates.
- Hardened the platform with OTP authentication (rotating refresh tokens, replay detection, rate limiting), provider-selectable payment gateways behind **HMAC-verified webhooks**, **440+ automated tests** (unit, PostGIS integration, Playwright end-to-end) and Prometheus/Grafana observability on a Dockerized cloud deployment.

## EDUCATION

**Bachelor of Technology (B.Tech), Information Technology** Dec 2021 - Jun 2025

University School of Information, Communication and Technology (USICT), GGSIP University, Delhi **– CGPA: 8.55/10**

## CERTIFICATIONS & ACHIEVEMENTS

**Certification – Postman API Fundamentals:** API design, documentation and automated testing workflows.
**Achievement – LeetCode:** Solved **800+** data structures and algorithms problems across arrays, strings, graphs, trees and algorithm design.
**Achievement – Adobe GenSolve:** Placed in the **top 5 percentile** nationally in the algorithmic coding competition.